

Axis-Angle
 $\omega @ \hat{\mathbf{n}}$

$SO(3)$
 $R_{ij} = \begin{pmatrix} \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{pmatrix}$

Eulerian
Euler Angles
Bunge: $(\varphi_1, \Phi, \varphi_2)$

Neo-Eulerian $\hat{\mathbf{n}} f(\omega)$

Stereographic
 $\mathbf{s} = \hat{\mathbf{n}} \tan \frac{\omega}{4}$

Equal-Angle

Equal-Volume

Rodrigues-Frank
 $\mathbf{r} = \hat{\mathbf{n}} \tan \frac{\omega}{2}$

Homochoric
 $\mathbf{h} = \hat{\mathbf{n}} \left(3(\omega - \sin \omega)/4 \right)^{1/3}$

$\mathbb{S}^3$
 $\mathbf{q} = \hat{\mathbf{n}} \sin \frac{\omega}{2}$
Quaternion
 $q = [q_0, \mathbf{q}]$

Equal-Volume

Rotation
 $\mathbf{v} = \omega \hat{\mathbf{n}}$

Cubochoric
 $\mathbf{c}$